# H524 Introduction to Biostatistics
## Public Health Data Analysis Project
## Fall 2025

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**Final Assessment Project Overview**

Teams of 7 students will conduct a comprehensive biostatistical analysis of a real public health dataset as the culminating course assessment (25% of final grade). This project replaces traditional final exams and demonstrates biostatistical analysis skills while maintaining scientific rigor and reproducibility.

# 1 Project Timeline

- **Week 3:** Project introduction and team formation
- **Week 4:** Dataset selection and research question formulation
- **Week 5:** Project proposal due (1 page)
- **Week 7:** Mid-project check-in and feedback
- **Week 9:** Draft analysis due for peer review
- **Week 10:** Final project submission
- **Finals Week:** Final assessment presentations (20 minutes + 5 minutes Q&A per team)

# 2 Available Datasets

Each team must select one of the following real public health datasets:

## 2.1 Option 1: CDC PLACES Health Data (County-Level)

- **Source:** CDC PLACES Project 2022
- **Focus Areas:** Chronic disease prevalence, health behaviors, prevention measures
- **Sample Size:** 3,100 counties/county equivalents
- **Access:** https://chronicdata.cdc.gov/browse?category=500+Cities+%26+Places
- **Data Type:** Aggregated county-level estimates (no survey weights needed)

- **Suggested Research Questions:**
  - Geographic patterns in diabetes prevalence across US counties
  - Association between preventive care access and health outcomes
  - Rural vs urban differences in chronic disease burden

### 2.2 Option 2: COVID-19 Case Surveillance Public Use Data

- **Source:** CDC COVID-19 Case Surveillance
- **Focus Areas:** COVID-19 outcomes, risk factors, demographics
- **Sample Size:** 1 million de-identified cases
- **Access:** `https://data.cdc.gov/Case-Surveillance/`
- **Suggested Research Questions:**
  - Risk factors for severe COVID-19 outcomes
  - Geographic and temporal patterns in case fatality rates
  - Age-stratified analysis of hospitalization predictors

### 2.3 Option 3: Heart Disease Mortality Data (County-Level)

- **Source:** CDC WONDER Underlying Cause of Death
- **Focus Areas:** Heart disease mortality rates, demographic patterns
- **Sample Size:** 3,100 counties with 5+ years of data
- **Access:** `https://wonder.cdc.gov/controller/datarequest/D77`
- **Data Type:** Aggregated mortality rates (no survey weights needed)
- **Suggested Research Questions:**
  - Geographic disparities in heart disease mortality
  - Age and gender patterns in cardiovascular death rates
  - Trends in heart disease mortality over time

### 2.4 Option 4: Global Burden of Disease (GBD) Study

- **Source:** IHME GBD 2019
- **Focus Areas:** Disease burden, mortality, risk factors globally
- **Sample Size:** 204 countries and territories
- **Access:** `http://ghdx.healthdata.org/gbd-results-tool`
- **Suggested Research Questions:**
  - Trends in disease burden across income levels
  - Environmental risk factors and respiratory disease mortality
  - Progress toward health-related SDG targets

### 2.5 Option 5: Youth Tobacco Use Data

- **Source:** CDC Youth Risk Behavior Surveillance System (State-Level)
- **Focus Areas:** Tobacco use patterns, demographics, risk behaviors
- **Sample Size:** 50 states + DC, multiple years
- **Access:** https://www.cdc.gov/healthyyouth/data/yrbs/
- **Data Type:** State-level aggregated percentages (no survey weights needed)
- **Suggested Research Questions:**
  - State-level variations in youth tobacco use
  - Trends in e-cigarette vs traditional cigarette use
  - Association between tobacco policies and use rates

### 2.6 Option 6: OpenFDA Adverse Events Database

- **Source:** FDA Adverse Event Reporting System (FAERS)
- **Focus Areas:** Drug safety, adverse events, pharmacovigilance
- **Sample Size:** Millions of reports
- **Access:** https://open.fda.gov/data/faers/
- **Suggested Research Questions:**
  - Patterns in adverse events for specific drug classes
  - Age and gender differences in drug safety profiles
  - Time trends in reporting of serious adverse events

## 3 Project Components

### 3.1 Part 1: Research Question and Literature Review (15%)

- Formulate a clear, answerable research question
- Identify gaps your analysis will address

### 3.2 Part 2: Data Management and Exploration (20%)

- Exploratory data analysis with appropriate visualizations
- Missing data assessment and handling strategy

### 3.3 Part 3: Statistical Analysis (30%)

- Apply appropriate statistical methods from H524 curriculum:
  - Descriptive statistics and data visualization
  - Hypothesis testing (t-tests, chi-square, ANOVA)
  - Correlation and regression analysis
  - Appropriate adjustments for multiple comparisons
- Conduct sensitivity analyses where appropriate

### 3.4 Part 5: Final Presentation and Communication (25%)

- 20-minute team presentation + 5 minutes Q&A
- Clear visual aids and professional data visualizations
- Effective storytelling with data and public health implications
- Demonstration of team collaboration and individual contributions
- Professional delivery, time management, and audience engagement

## 4 Deliverables

### 4.1 Project Proposal (Week 5)

One-page proposal including:

- Research question and hypothesis
- Selected dataset and justification
- Planned statistical approaches
- Team member roles and responsibilities

### 4.2 Final Report (Week 10)

Comprehensive report (12-15 pages) including:

- Executive summary (1 page)
- Introduction and literature review
- Methods section with data description
- Results with tables and figures
- Discussion and limitations
- Reproducible R code with comments
- Team contribution statement

### 4.3 Final Assessment Presentation Materials (Finals Week)

- PowerPoint/PDF slides (20-25 slides for 20-minute presentation)
- Live demonstration of key findings and statistical methods
- Prepared responses for anticipated questions and statistical challenges
- One-page executive summary handout for audience
- Individual reflection on learning (1 page per student)

## 5 Grading Rubric

| Component | Weight | Points |
|---|---|---|
| Research Question & Literature Review | 15% | 15 |
| Data Management & Exploration | 20% | 20 |
| Statistical Analysis | 25% | 25 |
| AI Integration & Documentation | 15% | 15 |
| Final Presentation & Communication | 25% | 25 |
| **Total** | **100%** | **100** |

### 5.1 Detailed Grading Criteria

#### 5.1.1 Excellent (90-100%)

- Clear, innovative research question with public health relevance
- Comprehensive statistical analysis with appropriate methods
- Exceptional AI integration with thorough documentation
- Professional presentation with compelling data story
- All team members contributed meaningfully

#### 5.1.2 Good (80-89%)

- Well-defined research question with clear objectives
- Appropriate statistical methods correctly applied
- Good AI usage with adequate documentation
- Clear presentation with good visualizations
- Most team members actively participated

#### 5.1.3 Satisfactory (70-79%)

- Basic research question addressed
- Statistical methods mostly correct with minor errors
- AI used but documentation incomplete
- Adequate presentation meeting time requirements
- Uneven team contribution evident

#### 5.1.4 Needs Improvement (¡70%)

- Unclear research question or objectives
- Inappropriate statistical methods or major errors
- Minimal AI usage or poor documentation
- Presentation lacks clarity or professionalism
- Limited team collaboration

# 6 Team Formation and Roles

Each team of 7 students should designate:

- **Project Manager:** Coordinate meetings, deadlines, and submissions
- **Data Manager:** Lead data cleaning and preprocessing
- **Statistical Lead:** Guide analytical approach and methods
- **AI Integration Lead:** Coordinate AI tool usage and documentation
- **Visualization Lead:** Create figures and presentation materials
- **Writing Lead:** Coordinate report writing and editing
- **Presentation Lead:** Coordinate final presentation and Q&A preparation

Note: All team members are expected to contribute to all aspects; roles are for coordination.

# 7 Resources and Support

## 7.1 Technical Resources

- RStudio with GitHub Copilot enabled
- Access to OSU library databases
- Canvas discussion board for project Q&A
- Sample analysis templates on Canvas

### 7.2 Office Hours Support

- Regular office hours: Wednesday 10:00-11:30 AM
- Special project office hours: Weeks 5, 7, 9
- Schedule team consultations via Canvas

### 7.3 Recommended R Packages

```
# Data manipulation
library(tidyverse)
library(data.table)

# Statistical analysis
library(stats)
library(car)
library(lmtest)
library(tableone)

# Visualization
library(ggplot2)
library(plotly)
library(corrplot)

# Reporting
library(knitr)
library(rmarkdown)
library(stargazer)
```

## 8 Academic Integrity

**Important Reminders:**

- All AI-generated content must be verified and attributed
- Each team member must contribute substantively
- Plagiarism or fabrication of data will result in project failure
- Inter-team collaboration is encouraged, but each team's work must be unique
- Citation of all sources (including AI tools) is required

## 9 Tips for Success

1. **Start Early:** Data cleaning always takes longer than expected
2. **Communicate Regularly:** Schedule weekly team meetings

3. **Document Everything:** Keep detailed notes of decisions and methods
4. **Verify AI Output:** Always double-check AI-generated code and interpretations
5. **Practice Presentation:** Rehearse timing and transitions
6. **Seek Feedback:** Use office hours and peer review opportunities
7. **Focus on Story:** Ensure your analysis tells a coherent public health story